

PROFESSIONAL SUMMARY

Environmental Data Analyst and Database Administrator with over 10 years of experience managing, validating, analyzing, and optimizing large-scale environmental, hydrological, meteorological, and water quality datasets. I am currently pursuing a Master's degree in Data Analytics and Artificial Intelligence with a focus on advanced analytics, machine learning, and environmental data systems. Experienced in data stewardship, data governance, quality assurance, relational database systems, and environmental data lifecycle management within regulatory and scientific environments. Strong background in SQL, Oracle, Microsoft SQL Server, MySQL, Python, Power BI, Tableau, and environmental data validation tools. Skilled in identifying and resolving data anomalies, supporting regulatory compliance, developing data validation workflows, and maintaining high-quality environmental records for operational and scientific programs.

CORE COMPETENCIES

- Environmental Data Stewardship and Data Governance
- Environmental Data Validation and Quality Assurance (QA/QC)
- Hydrological, Meteorological, and Water Quality Data Management
- Relational Database Administration (Oracle, SQL Server, MySQL)
- SQL Programming, Query Development, and Data Manipulation
- Environmental Data Lifecycle Management and Archival
- Data Integration and Electronic Data Deliverables (EDDs)
- Data Visualization and Reporting (Power BI, Tableau, Excel)
- GVA (Graphical Verification Analysis) and DBHYDRO Systems
- Data Anomaly Detection, Investigation, and Resolution
- Microsoft Excel (Advanced Functions, Macro-enabled Spreadsheets)
- Microsoft Access Database Management
- Metadata Documentation and Data Dictionary Support
- Python Data Processing and Automation
- Technical Support and User Training
- SAS Viya
- Databricks

EDUCATION

- Master's Degree in **Data Analytics and Artificial Intelligence**, 2026, Florida Gulf Coast University
- Bachelor of Applied Science (BAS) Degree in **Information Management – Database Administration and Data Analytics**, Palm Beach State College, Florida
- Associate of Science Degree in **Marketing Management & Entrepreneurship**, Miami Dade College, Florida
- Certificate in Introductory Computer Science, AFCOR Computer Training Center, Addis Ababa

EXPERIENCE

Environmental Data Analyst Galt Group INC / SFWMD's Hydro Data Management, Data Validation Unit, West Palm Beach, FL

July 2025 – Present

- Perform advanced environmental data validation and quality assurance for hydrological and

meteorological datasets from over 190 monitoring stations using GVA (Graphical Verification Analysis).

- Analyze, validate, reconcile, and maintain environmental datasets to ensure accuracy, completeness, consistency, and compliance with District standards and regulatory requirements.
- Utilize DBHYDRO, GVA, Data Depot, SQL, and database tools to compare and verify multiple environmental data sources.
- Detect, investigate, document, and resolve environmental data anomalies, telemetry issues, and inconsistencies through collaboration with engineers, scientists, and SCADA technicians.
- Develop and maintain detailed data quality annotations, metadata documentation, and validation records supporting audit compliance and long-term data integrity.
- Support environmental data lifecycle management including data acquisition, validation, reporting, and archival activities.
- Generate graphical analyses, dashboards, and reports supporting environmental monitoring, operational planning, and scientific decision-making.
- Apply statistical and graphical methods to analyze rainfall distribution, hydrologic trends, and water-level patterns supporting environmental modeling and operations.
- Provide technical support and guidance to internal users regarding environmental data retrieval, validation, and interpretation.

Database Administrator and Data Analyst, Ecohydro Consulting LLC,

October 2014 -July 2025, Boynton Beach, FL

- Designed, developed, and administered environmental and hydrological database systems using Oracle, Microsoft SQL Server, and MySQL.
- Led development of integrated Meteorological, Hydrological, and Environmental Data Management Systems supporting environmental studies and hydrological model development.
- Developed SQL queries, automated workflows, and data integration procedures for processing large environmental datasets from multiple sources.
- Conducted environmental data validation, quality checks, and data reconciliation to ensure data accuracy and consistency.
- Developed Power BI, Tableau, and Excel dashboards for environmental monitoring, reporting, and project performance tracking.
- Supported environmental data governance activities including metadata management, documentation, and data process standardization.
- Assisted with data preparation, reporting, and data exports for scientific studies and consulting projects.
- Trained staff on database systems, SQL querying, data visualization, and environmental data management best practices.
- Collaborated with technical teams on data integration, reporting systems, and database optimization projects.

Database Administrator, Ecohydroecom INC,

March 2017 - July 2025, Boynton Beach, FL, (parttime)

- Designed and maintained relational database systems supporting eCommerce operations using SQL Server, MySQL, and Oracle Database.
- Developed SQL queries, reporting systems, and dashboards for customer, sales, and inventory analytics.
- Managed database backup, recovery, indexing, and performance optimization activities.
- Built centralized repositories integrating customer, transaction, and website analytics data.
- Developed Power BI and Tableau dashboards supporting operational and business decision-making.
- Supported ASP.Net and SQL-driven backend systems integration projects.

KEY PROJECT ACCOMPLISHMENTS

Integrated Data Analytics and Machine Learning for Environmental Time-Series Analysis

- Developed a framework for environmental and hydrologic time-series analysis using statistical and machine learning methods including Linear Regression, KNN, and Random Forest.
- Applied data preprocessing techniques including anomaly detection, outlier identification, gap analysis, and automated gap filling using Python.
- Improved short-term environmental forecasting and supported data-driven environmental analysis through visualization and predictive modeling.

Meteorological, Hydrological, and Environmental Data Management System

- Designed and implemented an integrated environmental data management platform supporting large-scale hydrological and meteorological datasets.
- Performed environmental data preprocessing using Python, SQL, and ArcGIS for hydrological model development using SWAT.
- Built Power BI dashboards supporting environmental monitoring and decision-making.

TECHNICAL SKILLS

Database Systems

- Oracle Database
- Microsoft SQL Server
- MySQL
- Microsoft Access

Programming & Data Tools

- SQL
- Python (Pandas, NumPy)
- SAS Viya
- Visual Basic
- C#

Data Visualization & Reporting

- Power BI
- Tableau
- Microsoft Excel (Advanced Functions, Pivot Tables, Charts, Macros)

Environmental & Data Systems

- GVA (Graphical Verification Analysis)
- DBHYDRO
- Data Depot
- SWAT (Soil and Water Assessment Tool)

Publication:

Senait A Gebriye, Sedat Cevikparmak, Integrated Data Analytics and Machine Learning for Environmental Time-Series Analysis: Application to Hydrologic Data in South Florida, Poster CSDMS Annual Meeting, May 19–21, 2026, the University of Minnesota, Minneapolis